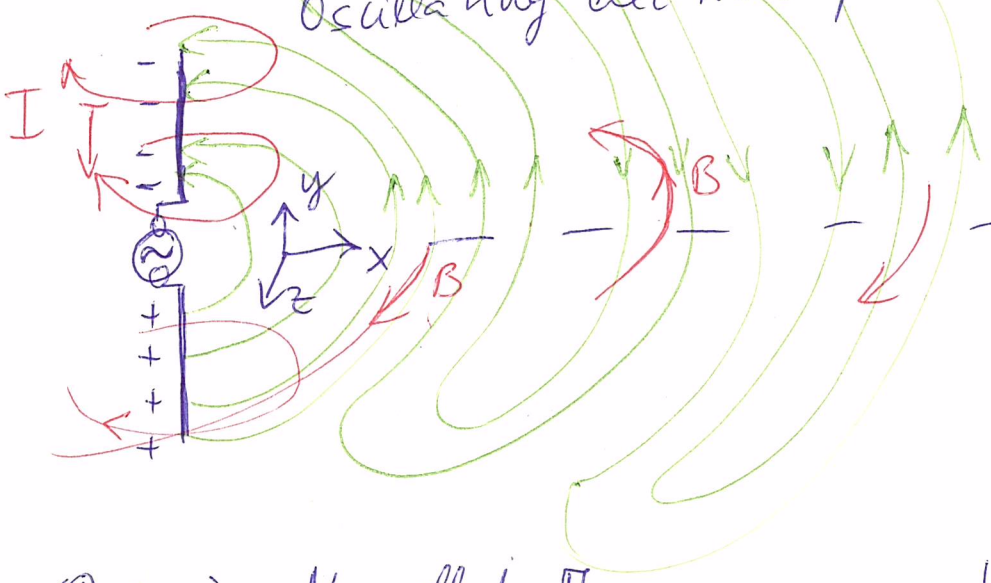
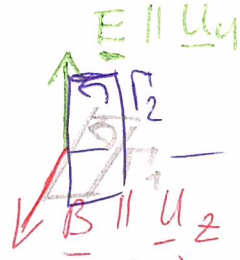
* Video

14. Electro-magnetic waves

Oscillating electric dipole (Hertz dipole) 1887



$\underline{E} = \begin{pmatrix} 0 \\ E_y \\ 0 \end{pmatrix}$



$\underline{B} = \begin{pmatrix} 0 \\ 0 \\ B_z \end{pmatrix}$

① Ampère Maxwell for Γ_1

$$\oint_{\Gamma_1} \underline{B}\,d\underline{r} = \mu_0\varepsilon_0 \iint_{A_1} \frac{\partial \underline{E}}{\partial t}\,d\underline{s} + \mu_0 I$$

$$B_z(x)\cdot l + 0 - B_z(x+\Delta x)\,l + 0 = \mu_0\varepsilon_0\, l\cdot\Delta x \frac{\partial E_y}{\partial t}$$

$$\frac{B_z(x) - B_z(x+\Delta x)}{\Delta x} = \mu_0\varepsilon_0 \frac{\partial E_y}{\partial t}$$

$$\frac{\partial B_z}{\partial x} = -\mu_0\varepsilon_0 \frac{\partial E_y}{\partial t} \quad / \frac{\partial}{\partial x}$$

$$\frac{\partial^2 B_z}{\partial x^2} = -\mu_0\varepsilon_0 \frac{\partial^2 E_y}{\partial t\,\partial x}$$

Faraday Γ_2

$$\oint_{\Gamma_2} \underline{E}\,d\underline{r} = -\iint_{A_2} \frac{\partial \underline{B}}{\partial t}\,d\underline{s}$$

$$-E_y(x)\cdot l + E_y(x+\Delta x)\,l = -l\Delta x \frac{\partial}{\partial t}$$

$$\frac{\partial E_y}{\partial x} = -\frac{\partial B_z}{\partial t} \Big/ \frac{\partial}{\partial t}$$

$$\frac{\partial E_y}{\partial x\,\partial t} = -\frac{\partial^2 B_z}{\partial t^2} \quad / \frac{\partial}{\partial x}$$

$/\frac{\partial}{\partial t}$

$$\frac{\partial^2 B_z}{\partial x^2} = (--)\,\mu_0\varepsilon_0 \frac{\partial^2 B_z}{\partial t^2}$$

$$\frac{\partial^2 B_z}{\partial x^2} - \mu_0\varepsilon_0 \frac{\partial^2 B_z}{\partial t^2} = 0$$

$$\frac{\partial^2 E_y}{\partial x^2} = -\frac{\partial^2 B_z}{\partial x \partial t}$$

$$\frac{\partial^2 B_z}{\partial x \partial t} = -\mu_0 \varepsilon_0 \frac{\partial^2 E_y}{\partial t^2}$$

$$\frac{\partial^2 E_y}{\partial x^2} = -\frac{\partial^2 B_z}{\partial t \partial x}$$

$$\frac{\partial^2 E_y}{\partial x^2} - \mu_0 \varepsilon_0 \frac{\partial^2 E_y}{\partial t^2} = 0$$

Wave equation for E_y and for B_z. → Transversal Wave

Solution: Wave function $E_y := \hat{E}_0 \cdot \sin(kx - \omega t)$

Substitute:

$$-k^2 E_0 \sin(kx - \omega t) + \mu_0 \varepsilon_0 \omega^2 E_0 \cdot \sin(kx - \omega t) = 0$$

$$E_0 \cdot \underbrace{(\omega^2 \mu_0 \varepsilon_0 - k^2)}_{=0} \sin(kx - \omega t) = 0 \quad \forall x, t$$

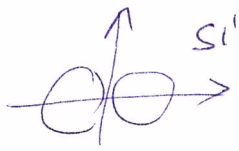
($E_0 \neq 0$, $\sin(kx - \omega t) \neq 0$)

$$\frac{\omega^2}{k^2} = \frac{1}{\mu_0 \varepsilon_0} \qquad v_{phase} = \frac{\omega}{k} = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3 \cdot 10^8 \frac{m}{s}$$

At the time of Maxwell they could measure the speed of light. (1676: $2{,}2 \cdot 10^8$ Roemer, Huygens
1850 $3.15 \cdot 10^8 \frac{m}{s}$ Fizeau
1862 2.98 Foucault)

So Maxwell derived the wave equation, foreseeing the existence of EM waves and suspecting light to be an EM wave. Heinrich Hertz proved experimentally in 1887

* Video



- No radiation ∥ to de
- E is oscillating in the plane of $\underline{de}$, $\underline{r}$
- B is -∥- $\perp \underline{E}, \underline{r}$

Differential description of EM waves

AM

$$\text{curl } \underline{H} = \frac{\partial \underline{D}}{\partial t} + \underline{j} \qquad \frac{\partial}{\partial t}$$

$$\text{curl } \frac{\partial \underline{H}}{\partial t} = \frac{\partial^2 \underline{D}}{\partial t^2} + \frac{\partial \underline{j}}{\partial t}$$

$$\frac{\partial^2 \underline{P}}{\partial t^2} = \varepsilon \frac{\partial^2 \underline{E}}{\partial t^2}$$

if ε const in time

Faraday

$$\text{curl } \underline{E} = -\frac{\partial \underline{B}}{\partial t} \quad / \text{curl}$$

$$\text{curl}(\text{curl } \underline{E}) = -\text{curl} \frac{\partial \underline{h}}{\partial t} = -\text{curl}\left(\mu \frac{\partial \underline{H}}{\partial t}\right)$$

$$\text{curl}(\text{curl } \underline{E}) = -\mu \, \text{curl} \frac{\partial \underline{H}}{\partial t}$$

if μ const in space

$$\text{curl}(\text{curl}(\underline{E})) = -\mu\varepsilon \frac{\partial^2 \underline{E}}{\partial t^2} - \mu \frac{\partial \underline{j}}{\partial t}$$

$$\text{grad}(\text{div } \underline{E}) - \Delta \underline{E} = \mu\varepsilon \frac{\partial^2 \underline{E}}{\partial t^2} - \mu \frac{\partial \underline{j}}{\partial t}$$

$$\boxed{\Delta \underline{E} - \mu\varepsilon \frac{\partial^2 \underline{E}}{\partial t^2} = \underbrace{\text{grad}(\text{div } \underline{E})}_{**} + \mu \frac{\partial \underline{j}}{\partial t}}$$

** ∅ if homogenous dielectrics

* source & dissipation ∅ in diel.

So in vacuum and homogenous lin. dielectrics:

$$\Delta \underline{E} - \mu\varepsilon \frac{\partial^2 \underline{E}}{\partial t^2} = 0$$

We are now in 3D! in material!

Simplest solution: plane wave (abstraction!)

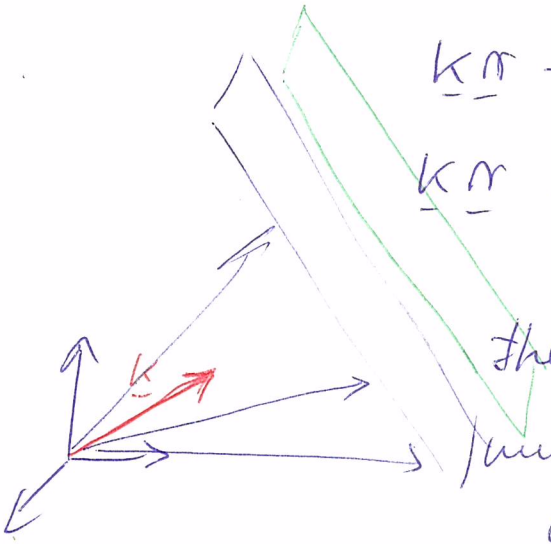
$$\underline{E} = \underline{E}_0 \cdot \cos(\underline{k}\,\underline{r} - \omega t)$$

Wave front: location of points having const phase

$$\underline{k}\,\underline{r} - \omega t = \text{const}_0 = \varphi_0$$

$$\underline{k}\,\underline{r} = \omega t_0 + \varphi_0 \quad \text{for a given } t_0$$

$= \text{konst}$



the projection of all $\underline{r}$ vectors ↑ to $\underline{k}$ must be const.

↙

a plane $\perp$ to $\underline{k}$

$t_1 > t_0$

$$(\underline{k}\,\underline{r})_1 = \omega t_1 + \varphi_0 = \text{const}_1 > \text{const}_0$$

The wave-front moves forward parallel to $\underline{k}$ in the direction of $\underline{k}$:

$$kr = \omega t_0 + \varphi_0$$

$$r = \underbrace{\frac{\omega}{k}}\, t_0 + \varphi_0$$

v_{phase}, $\underline{v}_{ph} \parallel \underline{k}$

Substitute $E_0 \cdot \cos(\underline{k}\underline{r} - \omega t)$

$$\Delta \underline{E} = \begin{pmatrix} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) E_x \\ (\quad\quad\quad\quad) E_y \\ (\quad\quad\quad\quad) E_z \end{pmatrix} = \begin{pmatrix} (-k_x^2 - k_y^2 - k_z^2) E_x \\ E_y \\ E_z \end{pmatrix}$$

$$= k^2 \cdot \underline{E}$$

$$\frac{\partial^2 \underline{E}}{\partial t^2} = -\omega^2 \underline{E} \quad \Rightarrow \quad -k^2 \underline{E} + \mu \varepsilon \omega^2 \underline{E} = 0 \quad \forall r, t$$

$$\Downarrow$$

$$\frac{\omega}{k} = \frac{1}{\sqrt{\varepsilon \mu}} = \frac{1}{\sqrt{\varepsilon_0 \mu_0}} \cdot \frac{1}{\sqrt{\varepsilon_{rel} \mu_{rel}}}$$

$$\sqrt{\varepsilon_0 \mu_0} \rightsquigarrow c = 3 \cdot 10^8 \frac{m}{s} \qquad \mu_{rel} \approx 1$$

$$\varepsilon_{rel} \geq 1$$

$$v_{ph} = \frac{\omega}{k} = c/n \qquad n \text{ is typically} > 1$$

EM waves propagate slower in matter !

We forgot about $\underline{B}$. We can derive a wave equation for B, but we have a simpler solution:

Apply Faraday's law again:

$$\text{curl}\,\underline{E} = -\frac{\partial \underline{B}}{\partial t} \qquad E = E_0 \cdot \cos(k_x x + k_y y + k_z z - \omega t)$$

$$\text{curl}\,\underline{E} = \underline{\nabla} \times \underline{E} = \begin{pmatrix} \partial_y E_z - \partial_z E_y \\ \partial_z E_x - \partial_x E_z \\ \partial_x E_y - \partial_y E_x \end{pmatrix} =$$

$$= -\begin{pmatrix} k_y E_{0z} - k_z E_{0y} \\ k_z E_{0x} - k_x E_{0z} \\ k_x E_{0y} - k_y E_{0x} \end{pmatrix} \cdot \sin(\underline{k}\underline{r} - \omega t) =$$

$$= -\underline{k} \times \underline{E}_0 \cdot \sin(\underline{k}\underline{r} - \omega t)$$

in complex notation
$E_0 e^{i(\underline{k}\underline{r} - \omega t)}$
$\nabla \times \underline{E} = i\underline{k} \times \underline{E}$

Let's suppose that the "magnetic wave" has different $\underline{k}', \omega', \varphi_0$ $\qquad \underline{B} = \underline{B}_0 \cos(\underline{k}'\underline{r} - \omega' t + \varphi_0)$

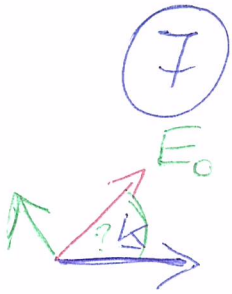
$$-\frac{\partial \underline{B}}{\partial t} = -\omega' \underline{B} \sin(\underline{k}'\underline{r} - \omega' t + \varphi_0)$$

$$-\underline{k} \times \underline{E}_0 \sin(\underline{k}\underline{r} - \omega t) = -\omega' B_0 \sin(\underline{k}'\underline{r} - \omega' t + \varphi_0) \qquad \forall \underline{r}, t$$

there is no separate electric & magnetic wave! They are coupled, they propagate in the same direction, same λ, same phase!

$\underline{k}' = \underline{k}$
$\omega' = \omega$
$\varphi_0 = 0$

$$-\underline{k}\times\underline{E}_0 = -\omega\underline{B}_0 \Rightarrow \underline{B}_0 = \frac{\underline{k}\times\underline{E}_0}{\omega}$$



B_0 is $\perp$ to $\underline{k}$ & $\underline{E}$

We suspect that E is also orthogonal: $|B_0| = \frac{|E_0|}{\omega}|\underline{k}| = E_0 \frac{1}{V_{ph}}$ ✓

We can apply A-M: $\operatorname{curl}\underline{H} = \frac{\partial \underline{D}}{\partial t} + \underline{j}$

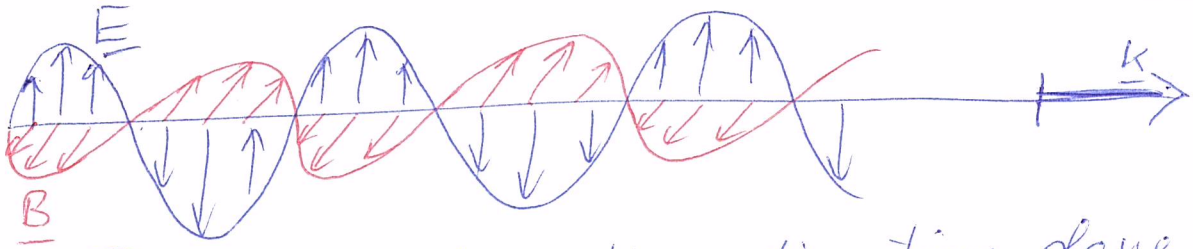
$$\Downarrow$$

$$\underline{D} = -\frac{\underline{k}\times\underline{H}}{\omega}$$

for a hom. isotropic lin. material:

$$\varepsilon\underline{E} = -\frac{\underline{k}\times\underline{B}/\mu}{\omega}$$

$$\underline{E} = -\frac{\underline{k}\times\underline{B}}{\omega}\cdot\frac{1}{\varepsilon\mu} \qquad \left(|E| = \frac{|B|}{V_{ph}}\cdot V_{ph}^2 \checkmark = |B|\cdot V_{ph}\right)$$



Def: the polarization ~~direction~~ plane of an EM wave is the plane of $\underline{E}$ (E is not more important, $\underline{E}$ and $\underline{B}$ have equal rank, it is simply a convention)

* What happens if $\underline{j} \neq 0$ on the right of the wave eqn.?

$$\Delta \underline{E} - \mu \varepsilon \frac{\partial^2 \underline{E}}{\partial t^2} = \mu \frac{\partial \underline{j}}{\partial t}$$

This term will describe the generation and absorption of waves

Source

To make an EM wave we have to have a changing current $\equiv$ accelerating charge. If we have an accelerating charge, it will radiate! (Bremsstrahlung) (see e.g. Hertz dipole!)

It is quite simple to see what happens to the EM wave in a conductive material:

$$\underline{j} = \tilde{\sigma} \underline{E}$$

$$\Delta \underline{E} - \mu \varepsilon \frac{\partial^2 \underline{E}}{\partial t^2} = \mu \frac{\partial \underline{j}}{\partial t} = \mu \tilde{\sigma} \frac{\partial \underline{E}}{\partial t}$$

test function

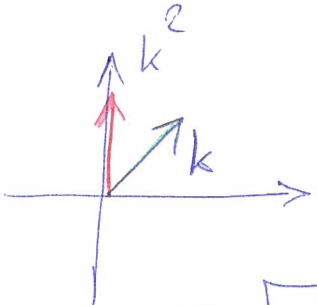
$$\underline{E} := \underline{E}_0 \cdot e^{i(\underline{k}\underline{r} - \omega t)}$$

$$-k^2 \underline{E} + \mu \varepsilon \omega^2 \underline{E} = -i\omega \mu \tilde{\sigma} \underline{E}$$

$$k^2 = \mu \varepsilon \omega^2 + i\omega \tilde{\sigma} \mu$$

$$= \mu\omega \left(\varepsilon \omega + i \tilde{\sigma} \right)$$

10^{-12} 10^{14} 10^{15}



$$\mathrm{Im}(k) = \sqrt{\frac{\omega \tilde{\sigma} \mu}{2}}$$

SKIN Effect!

III Note: if the medium is not homogenous:

** $\Delta \underline{E} - \mu\varepsilon \frac{\partial^2 \underline{E}}{\partial t^2} = \dots \quad grad(div\,\underline{E}) = grad\left(div \frac{\underline{D}}{\varepsilon}\right) =$

$= grad\left[\frac{div\,\underline{D}\cdot\varepsilon - \underline{D}\,grad(\varepsilon)}{\varepsilon^2}\right] = grad\left[\underbrace{\frac{div\,\underline{D}}{\varepsilon}}_{=\frac{\rho}{\varepsilon}=0} - \frac{\underline{D}\,grad\varepsilon}{\varepsilon^2}\right]$

$= -grad\frac{(\underline{D}\cdot grad\,\varepsilon)}{\varepsilon^2}$ ← if ε has a space variation,

it will act similarly to a source of radiation! Scattering of EM radiation in inhomogenous dielectrics

(fog!, snow, ...)